



SALAS O'BRIEN

3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Falcon Landing; D.R. Horton Atlanta West
Location: Gainesville, GA
Description: Mansfield - P, Slab

Date: September 22, 2024
By: TJH
House Faces: Worst West

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	3,326 ft ²
Volume	28,548 ft ³
Percentage Glass	9 %
Number of Bedrooms	5

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	10883
SEER	15
ACH50	5.0
Mechanical Ventilation	NO
House Outside Air CFM	N/A

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.067	Basement	Concrete + 2x6	R-13 batt	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.5	Exterior Door		R-2	
	D2	0.5	Garage Door		R-2	
Roofs	#	U-Value	Description/Notes			
	R1	0.026	Flat Ceiling	2x4	R-38 blown	
	R2	0.053	Vaulted		R-19 batt	
Floors	#	U-Value	Description/Notes			
	F1	0.053	Framed		R-19 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.35	Blinds/Draperies	SHGC 0.22
	G2					
	G3					

LOAD SUMMARY					
		Unit 1 - Whole House			
		Cooling		Heating	
Sensible Load		32819 Btu/hr		39335 Btu/hr	
Minimum Nominal Capacity		3.65 Tons		11.53 kW	
Load Density		912 SF/Ton		3.5 W/SF	
Floor Area Served		3326.0 SF		3326.0 SF	
Mechanical Ventilation		N/A CFM		N/A CFM	
System Size		4.0 Tons		15.0 kW	
Leaving Air Temperature		57.2 deg F		99.6 deg F	
Installed Capacity		832 SF/Ton		4.5 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

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[illegible]

Unit 1

Size

4.00

Tons

Airflow

1600

CFM

Heat kW

15

kW

First Floor Airflows

Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Foyer / Hall	54	78	66	
Powder	12	13	12	
Garage Entry	33	44	39	
Pantry	15	15	15	
Bed 5 WIC	14	20	17	
Bed 5	73	112	92	
Kitchen	169	75	122	
Breakfast	44	67	56	
Family	160	171	166	
Dining	29	45	37	
Living	91	116	104	
-	-	-	-	
-	-	-	-	
-	-	-	-	

Unit 1 (Continued)				
Second Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Stairs	20	37	28	
Hall	49	42	46	
Laundry	49	9	29	
Bath 3	16	23	19	
Bed 2 & WIC	127	110	118	
Rec Rm	211	147	179	
Bed 3	101	103	102	
Bed 3 WIC	18	28	23	
Bed 4	67	70	69	
Bed 4 WIC	15	27	21	
Bed 1 WIC	27	34	30	
Bath 1 & Closet	38	41	39	
Bed 1	156	162	159	
Bath 2	14	11	13	

Cooling Size	2	DX Cooling kWh	1770
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Heating Size	2	RC Heating kWh	3214
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Best Case	1.5	Total kWh	4984
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SEER 15

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Project:	Falcon Landing: D.R. Horton Atlanta West	Date:	September 22, 2024	Typical Ceiling Height	8	Cooling Size	2	DX Cooling kWh	2314
Location:	Gainesville, GA	By:	TJH			Heating Size	2	RC Heating kWh	3584
Description:	Mansfield - P, Slab	Level(s):	Second Floor	Kneewall Height	1	Best Case	2	Total kWh	5899
Square Feet		1976	Volume	16398	SF/Ton	988	SEER	15	

COOLING LOAD CALCULATION							Ceiling Height		18		8		8		8		8		8		8		8		8		8		8		8		8							
Components					CLTD (°F)	U-Value (Btu/hr-sf-F)	Cooling Load Factor	Entire House		Stairs		Hall		Laundry		Bath 3		Bed 2 & WIC		Rec Rm		Bed 3		Bed 3 WIC		Bed 4		Bed 4 WIC		Bed 1 WIC		Bath 1 & Closet		Bed 1		Bath 2				
Type	Description			Qty				Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr				
Glass	G1	South	Right	(E)		16	Btu/hr-sf	30	474		-		-		-		-		-		-		-		-		-		-		-	30	474		-					
	G1	SouthWest		(SE)		26	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-	-	-	-						
	G1	West	Front	(S)		29	Btu/hr-sf	75	2212		-		-		-		-	30	885	30	885	15	442		-		-		-		-	-	-	-						
	G1	NorthWest		(SW)		18	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-	-	-	-						
	G1	North	Left	(W)		6	Btu/hr-sf	15	93		-		-		-		-		-		-		-		15	93		-		-	-	-	-	-						
	G1	NorthEast		(NW)		8	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-	-	-	-						
	G1	East	Back	(N)		13	Btu/hr-sf	30	392		-		-		-		-		-		-		-		-		-		-		-	30	392		-					
	G1	SouthEast		(NE)		14	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-	-	-	-						
	G1	Sky Light				46	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-	-	-	-						
Door	D1	Door				6	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-	-	-	-						
	D2	Door			21	0.50	10.50	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-	-	-	-	-						
Above Grade Walls	W1	South			55	0.50	27.50	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-	-	-	-	-						
	W1	SouthWest			21	0.08	1.23	Btu/hr-sf	414	510	73	90	-	-	-	75	92	116	143	18	22	-	-	-	-	-	-	-	-	-	-	-	132	163	-	-				
	W1	West			23	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	NorthWest			19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	North			13	0.08	1.00	Btu/hr-sf	390	390	-	-	-	-	-	-	-	-	9	9	18	18	113	113	76	76	115	115	60	60	-	-	-	-	-					
	W1	NorthEast			19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	East			23	0.08	1.77	Btu/hr-sf	402	712	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	49	86	111	197	96	170	146	259	-	-				
	W1	SouthEast			21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	South			16	0.08	1.23	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
Basement Walls	W1	SouthWest			21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	West			23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	NorthWest			19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	North			13	0.08	1.00	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	NorthEast			19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	East			23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	SouthEast			21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	South			16	0.08	1.23	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	SouthWest			21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
Other	W2	Grade			0	0.07	0.00	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	Partition			10	0.08	0.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	Garage			30	0.08	2.31	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W3	Attic			55	0.05	2.92	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
Roof	W3	Kneewall			55	0.05	2.92	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	R1	Ceiling			55	0.03	1.43	Btu/hr-sf	1976	2826	59	84	207	296	58	83	52	74	216	309	308	440	164	235	46	66	189	270	36	51	78	112	140	200	353	505	70	100		
	R2	Vaulted			55	0.05	2.92	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-					
Floor	F1	Crawl/Uncon.			11	0.05	0.58	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-					
	F1	Garage			30	0.05	1.59	Btu/hr-sf	629	1000	-	-	72	114	-	-	-	-	308	490	164	261	46	73	39	62	-	-	-	-	-	-	-	-						
	F2	Slab			0	0.10	0.00	Btu/hr-sf	52	0	52	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
Internal	People						255	Btu/hr	5	1275	-	-	-	-	-	-	1	255	1	255	1	255	1	255	1	255	1	255	1	255	1	255	1	255	1	255				
	Lights (W)						3.413	Btu/hr-W	990	3379	30	102	104	355	29	99	26	89	108	369	154	526	82	280	23	78	95	324	18	61	39	133	70	239	177	604	35	119		
	Appliances (W)						3.413	Btu/hr-W	450	1536	-	-	-	200	683	-	-	-	-	250	853	-	-	-	-	-	-	-	-	-	-	-	-	-	-					
Vent	Infiltration				0.25	ACH	0.09	Btu/hr-cf	1476		-	-	96	149	42	37	-	-	156	222	-	-	-	-	33	136	26	56	101	254	50									
	Volume					CF		16398		1062		1656		464		416		1728		2464		1312		368		1512		288		624		1120		2824		560				
	Area					SF		1976		59		207		58		52		216		308		164		235		46		66		189		270		36		51				
Subtotal (Btu/hr)								16907		372		914		906		293		2359		3927		1885		326		1256		285		497		710		2905		270				
Safety Factor								10	%																															
Sensible Cooling Load (Btu/hr)								18597																																
Nominal Tons Required (75% sens. cap.)								2.07																																
System Selection								2 Tons		Space CFM		800		18		43		43		14		112		186		89		15		59		14		24		34		137		13
HEATING LOAD CALCULATION							Ceiling Height		18		8		8		8		8		8		8		8		8		8		8		8		8		8					
Components					CLTD (°F)	U-Value (Btu/hr-sf-F)	Heating Load Factor	Entire House		Stairs		Hall		Laundry		Bath 3		Bed 2 & WIC		Rec Rm		Bed 3		Bed 3 WIC		Bed 4		Bed 4 WIC		Bed 1 WIC		Bath 1 & Closet		Bed 1		Bath 2				
Type	Description			Qty																																				